

Quotients and Remainders with Models

Answer Key

Grade 4 — Whole-Number Division

Quick Answers

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|--|--|
| 1. (a) counters in each full group = 4, (b) counters left over = 3, — | 7. 8 |
| 2. (a) complete rows = 5, (b) tiles left over = 4 | 8. 7 |
| 3. (a) vans needed = 5 vans, — | 9. (a) 1235 rounded to the nearest hundred = 1200, (b) the estimated quotient = 200, (c) exact quotient = 205, (c) exact remainder = 5 |
| 4. 7 | 10. (a) pencils each friend gets = 13, (b) pencils left over = 5, (c) boxes needed to hold them all = 14, — |
| 5. (a) 356 rounded to the nearest hundred = 400, (b) the estimated quotient = 50, (c) exact quotient = 44, (c) exact remainder = 4 | |
| 6. 24 | |
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What is verified

7 of 10 problems fully machine-checked against SymPy.
 — marks an answer only you can judge — problems 1, 3, 10. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 4 — Whole-Number Division

4.NBT.B.6 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*

Difficulty 1–5 of 5 across 23 tagged checks.

Common wrong answers

3. *If they got 4: ignored the leftover students and reported only the number of full vans*
 4. *If they got 8: rounded the leftover cookies up into an eighth box that is not full*
 8. *If they got 5: answered with the number of full bags instead of the marbles left over*
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1. 23 counters in groups of 5. Circle the groups. (a) Full groups? (b) Left over?

The circled model — judged by eye. Look for four rings, each around exactly five counters, with three counters left outside every ring. Rings of different sizes mean the groups were not equal; a fifth ring drawn round the three leftovers is the error this problem is built to expose — a remainder is not a group.

Solution (a): Circling fives across the row gives four complete circles before the counters run out: 5, 10, 15, 20.

4

Solution (b): Four groups use $4 \times 5 = 20$ counters, and $23 - 20 = 3$ are left. The remainder is smaller than the group size, which is the check that no further group could be made. 3

2. 34 tiles in rows of 6. (a) Complete rows? (b) Tiles left over?

Solution (a): Count by sixes: 6, 12, 18, 24, 30 — five complete rows, and 36 would be too many. 5

Solution (b): $34 - 5 \times 6 = 34 - 30 = 4$, which matches the short row at the bottom of the model. 4

3. 38 students, vans seat 8. (a) Vans needed? (b) Why do the leftover students change the answer?

Solution (a): $38 \div 8 = 4$ with remainder 6. Four vans seat only 32 students, so a fifth van is needed for the remaining six. 5 vans

Solution (b) — instructor-judged. A full-credit answer says the remainder is six students, that those students still have to travel, and that a fifth van is therefore needed even though it will not be full. Model answer: “Dividing gives 4 remainder 6. Those 6 students cannot be left behind, so one more van is added, making 5.” Repeating the number five with no mention of the leftover students is not full credit.

4. 45 cookies, boxes of 6, only full boxes sold. How many full boxes?

Solution: $45 \div 6 = 7$ with remainder 3. Here the remainder is dropped, not rounded up: the three extra cookies do not make a full box, and only full boxes are sold. Compare with the van problem, where the same leftover forced an extra container — the arithmetic is identical, the decision is not. 7

5. 356 bulbs into 8 beds. (a) Round. (b) Estimate. (c) Exact.

Solution (a): The tens digit of 356 is 5, so it rounds up to the next hundred. 400

Solution (b): $400 \div 8 = 50$, so expect about fifty bulbs per bed. 50

Solution (c): $8 \times 40 = 320$, leaving 36; $8 \times 4 = 32$, leaving 4. So the quotient is $40 + 4 = 44$ and the remainder is 4. The exact answer sits just under the estimate, which is what rounding up predicts. 44

remainder

4

6. A bar of 96 counters split into 4 equal parts. How many in each part?

Solution: The bar model shows one whole of 96 cut into four equal pieces, so this is $96 \div 4$. Split it by place value: $80 \div 4 = 20$ and $16 \div 4 = 4$, so each part holds $20 + 4 = 24$. There is nothing left over — the parts really are equal. 24

7. 60 counters in 7 equal groups with 4 left over. How many in each group?

Solution: Read the model as an equation: $7 \times (\text{group size}) + 4 = 60$. Take the remainder off first: $60 - 4 = 56$ counters are actually in groups. Then $56 \div 7 = 8$. 8

8. 52 marbles into bags of 9. How many are left over?

Solution: $9 \times 5 = 45$ and $9 \times 6 = 54$ is too many, so five bags fill and $52 - 45 = 7$ marbles remain. The question asks for the leftovers, not the bags, so the answer is the remainder. 7

9. 1,235 leaflets into 6 cartons. (a) Round. (b) Estimate. (c) Exact.

Solution (a): The tens digit of 1235 is 3, below 5, so it rounds down to the nearest hundred. 1200

Solution (b): $1200 \div 6 = 200$, so expect about two hundred per carton. 200

Solution (c): $6 \times 200 = 1200$, leaving 35; $6 \times 5 = 30$, leaving 5. The quotient is $200 + 5 = 205$ and the remainder is 5. The estimate of 200 was close, as expected after rounding down by only 35. 205

remainder 5

10. 83 pencils, 6 friends / boxes of 6. (a) Each friend gets? (b) Left over? (c) Boxes needed? (d) Why do (a) and (c) differ?

Solution (a): $6 \times 13 = 78$ and $6 \times 14 = 84$ is too many, so each friend gets 13. 13

Solution (b): $83 - 78 = 5$. 5

Solution (c): Thirteen boxes hold 78 pencils and five are still loose, so a fourteenth box is needed. 14

Solution (d) — instructor-judged. A full-credit answer says the division was the same ($83 \div 6 = 13$ remainder 5) and only what happens to the remainder changed. Model answer: “Both times I got 13 remainder 5. When sharing pencils the 5 left over just are not handed out, so the answer is 13. When packing boxes those 5 still need somewhere to go, so I add a box and get 14.” Giving both numbers without saying what the remainder does in each case is not full credit.